

1.

(15 marks)

1.1 Assume that z is the complex number such that $|z| + 2\bar{z} - 3z = \frac{3i^{2560} - 45i^{2561}}{i^{128}}$.

Evaluate z^{-1} and $\left| \frac{z^{-1}z}{z^2} \right|$.

1.2 Let $|(4\sqrt{7} - 12)(-3 - \sqrt{7}i)z^6| = 8$. Find $z \cdot \bar{z}$.

2. Let x be the complex number. Find the roots of equation

$$x^7 - 8x^4 + 4x^3 - 20x^3 + 128 = 0.$$

(5 marks)

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3. Let z_0 , z_1 and z_2 be 3rd roots of $\sqrt{2}z = 1 + i$.
Find the value of $\frac{4z_0^4 + 3z_1^4}{2z_2^2}$ in the form of $a + bi$.

(10 marks)

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4. Let $(A - B)^T = \begin{bmatrix} 1 & 4 & -2 \\ 0 & 2 & 1 \\ -1 & 3 & 5 \end{bmatrix}$ and $2A = \begin{bmatrix} 6 \\ 8 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & -2 \end{bmatrix}$

where I is the identity matrix. Find the symmetric matrix (S) and skew-symmetric matrix (K) of B .

(10 marks)

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5. The function f is defined as determinant as the following

$$f(x) = \begin{vmatrix} e^x & e^x & e^x \\ e^{-3x} & e^{4x} & e^{5x} \\ e^{-5x} & e^{6x} & e^{7x} \end{vmatrix}.$$

- a. Find the value of $f(0)$. (4 marks)
b. Compute the value of $f(\ln 2)$. (6 marks)

Hint: Formula $e^{\ln x} = x$ and $\ln x^n = n \ln x$.

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6. Given two 3x3 square matrices $B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 2 & -3 \\ 3 & -1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$

which satisfy the following equation $AB - AC^T - \frac{1}{3}I = \underline{0}$, where I is a 3x3 identity matrix, $\underline{0}$ a 3x3 zero matrix and A a 3x3 square matrix. Compute $\text{adj}(A)$.

(10 marks)

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7. Give the system of linear equations

$$x_1 + x_2 + x_3 + x_3 + x_4 = 0$$

$$x_1 + x_2 + x_3 + x_3 - x_4 = 4$$

$$x_1 + x_2 - x_3 + x_3 + x_4 = -4$$

$$x_1 - x_2 + x_3 + x_3 + x_4 = 2.$$

Apply Gauss-Jordan method to solve the system and evaluate the value of $\frac{x_3^2 + x_4^2}{x_1^2 + x_2^2}$.

(10 marks)

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8. Solve the system of three equations of three variables

$$\frac{1}{x} + \frac{2}{y} + \frac{3}{z} = 6$$

$$\frac{1}{x} + \frac{3}{y} + \frac{5}{z} = 9$$

$$\frac{2}{x} + \frac{5}{y} + \frac{10}{z} = 16$$

By Cramer's rule and compute the value of x^{y^z} .

(10 marks)

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